Verified Localized Episodic Grounding with Counterexample Rule Refinement for Auditable LLM-Assisted Network Traffic Analysis

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Abstract: Memory-augmented large language models (LLMs) can assist few-shot network intrusion triage by summarizing historical flows into human-auditable rules without parameter updates, yet local reflections risk label hallucination, cross-flow over-generalization, and prompt-level anchoring on non-firing rules. We propose Verified Localized Episodic Grounding (VLEG) and its counterexample-refined neuro-symbolic extension (VLEG-CR-NS), an auditable consolidation and hybrid-inference framework for few-shot network traffic analysis. During consolidation, VLEG checks candidate reflections against their labeled source records within K attempts, after which a deterministic compiler parses admitted heuristics into executable Boolean predicates and feeds within-bank contrary-label counterexamples back to the generator to specialize overly broad rules. In a 16-scene symbolic sanity check, bounded source verification ($K = 3$) eliminates label inversions (100% observed agreement), performing comparably to raw retrieval at 70B (91.67% versus 89.58%). On a five-seed, 32-flow benchmark over 1000 deduplicated UNSW-NB15 test fingerprints, unverified single-step reflection mislabels 95.0% (76/80) of malicious support flows as benign (42.66% Macro-F1). Source-only VLEG restores 100% source-label agreement (158/158) and lifts Macro-F1 to 59.68%, but trails raw retrieval (63.92%) because 59.2% (93/157) of rules match within-seed counterexamples. Counterexample-guided refinement (VLEG-CR) raises zero-counterexample rule convergence from 40.0% to 84.4% (135/160) and cuts unseen cross-seed counterexamples from 72.0% to 40.7%. Executing compiled zero-counterexample rules deterministically before falling back to raw retrieval avoids LLM false alarms from non-firing rules. VLEG-CR-NS reaches 64.85% Macro-F1 (+0.93 percentage points over raw retrieval, 95% confidence interval [+0.13, +1.71]), while support-filtered bank triage (VLEG-CR-Bank) achieves 65.30% Macro-F1 (+1.38 percentage points, 95% confidence interval [+0.28, +2.60]) and the lowest false positive rate (35.99%) among all 32-flow baselines while eliminating 47.1%–49.0% of online LLM calls.

Keywords: Episodic memory; hallucination mitigation; large language models; network traffic analysis; intrusion detection; counterexample-guided refinement; neuro-symbolic execution

1 Introduction

1.1 Background and Motivation

Large language models (LLMs) can leverage demonstrations in the prompt to support classification and incident analysis without gradient updates [1, 2]. In network traffic analysis and Security Operations Center (SOC) intrusion triage [3–6], historical flow telemetry and incident summaries supply essential context for interpreting novel alerts. While supervised tabular classifiers such as Random Forest (RF), Gradient Boosting Decision Trees (GBDT), and Classification and Regression Trees (CART) provide high throughput when trained on tens of thousands of labeled records, analysts investigating scarce or emerging attack patterns require explicit, human-auditable rationales linking a target flow to historical precedents. Finite and unevenly utilized context windows make the consolidation and presentation of this historical evidence critical [7, 8].

Persistent-memory architectures extend experience beyond a single context window [9–13]. Retaining raw records preserves instance-level fidelity, whereas compressing episodes into natural-language rules offers

explicit decision guidance. However, an unverified summary can invert a class label, omit a discriminating condition, or turn an incidental feature value into an overly broad rule. Even when a rule is syntactically valid and faithful to its originating flow, injecting its text into an LLM prompt on target flows where its antecedent does not hold can induce false alarms through label anchoring.

Importantly, source agreement, cross-episode specificity, and execution fidelity are distinct requirements. Checking a proposed explanation against a single originating flow exposes direct label inversions or fabricated attributes, yet cannot reveal whether the stated condition falsely matches contrary-label traffic elsewhere in the support bank. Furthermore, once a rule is refined to eliminate contrary-label counterexamples, evaluating its numerical inequalities via deterministic symbolic execution differs fundamentally from asking a quantized LLM to interpret multiple conditional rules in text. Evaluating a memory system therefore requires tracking source-label consistency, within-seed and unseen cross-seed counterexample rates, deterministic rule coverage, and end-to-end classification metrics jointly.

1.2 Memory Research and Study Positioning

Memory-augmented language models reuse past observations through episodic retrieval, local reflection, and global semantic consolidation [14, 15]. Feedback-driven agents revise behavior across trials [16, 17], while global hypothesis learners induce abstractions across multiple examples [18, 19]. Our study focuses on a closed-loop formulation combining per-episode source verification, counterexample-guided inductive rule refinement, and neuro-symbolic rule triage, benchmarked directly against raw feature retrieval, global summaries, shallow CART decision trees, and tree ensembles.

Three research questions guide our investigation. The first question asks whether checking a local reflection against its labeled source episode eliminates the label hallucinations of unverified reflection, and how revision unfolds across verification iterations. The second question asks why 100% source-consistent reflections still inflate false positive rates on real network telemetry, and whether feeding within-seed counterexamples back to the generator can specialize rules without memorizing continuous float literals. The third question asks whether executing compiled zero-counterexample predicates deterministically prior to raw-retrieval fallback overcomes prompt-level anchoring on non-firing rules and outperforms raw retrieval under controlled retrieval budgets.

1.3 Proposed VLEG-CR and Neuro-Symbolic Triage Framework

We propose Verified Localized Episodic Grounding (VLEG), together with Counterexample-Refined consolidation (VLEG-CR) and Neuro-Symbolic triage (VLEG-CR-NS and VLEG-CR-Bank). During Phase 1 consolidation, a generator proposes a local reflection from label-free attributes, and a strict verifier checks it against the labeled source episode within budget K. Next, a deterministic compiler Ψ parses each admitted `if ... then ...` heuristic into an executable Boolean predicate over the 19 flow features and tests it against the remaining 31 historical flows in the same seed bank. Whenever a rule matches contrary-label flows, VLEG-CR returns the matched counterexamples and candidate separating feature bounds to the generator for up to K_{CR} refinement steps, admitting only rules that satisfy both $\mathrm{CE}_{\mathrm{seed}} = 0$ and source verification.

During Phase 2 inference, rather than unconditionally concatenating rule texts that may not apply to the target flow, VLEG-CR-NS evaluates compiled $\mathrm{CE}_{\mathrm{seed}} = 0$ predicates deterministically on the target feature vector in sub-millisecond CPU time. When firing rules agree on a class label, the system outputs the deterministic prediction paired with the firing human-readable rule(s) at zero LLM query cost; only uncovered or conflicting boundary flows invoke 70B LLM raw-neighbor inference.

1.4 Experimental Evaluation and Summary of Contributions

We evaluate our framework on a 16-scene symbolic sanity check at 7B and 70B scales and on deduplicated UNSW-NB15 network telemetry [37] spanning five 32-flow historical seeds, a disjoint 100-query pilot set, and 1000 formal test fingerprints per seed, alongside shallow CART trees, RF, and GBDT baselines. The main contributions of this work are four-fold:

- **Two-Stage Verified and Counterexample-Refined Memory Consolidation:** We formulate VLEG and VLEG-CR, coupling model-based source verification with a deterministic predicate compiler Ψ that eliminates within-seed counterexamples via iterative inductive specialization while preventing exact-equality memorization on continuous flow features.
- **Controlled Symbolic Sanity Check Across Model Scales:** Across five symbolic evaluations at 7B and 70B scales, bounded verification ($K = 3$) eliminates source-label inversions (100% observed agreement versus 41.7%–50.0% for unverified reflection) and matches raw retrieval accuracy at 70B (91.67% versus 89.58%), while isolating the theoretical boundary between single-source consistency and distractor suppression.
- **Mechanistic Diagnosis of Label Collapse, Over-Generalization, and Non-Firing Prompt Anchoring:** On UNSW-NB15, we show that unverified reflection mislabels 95.0% (76/80) of malicious flows as benign (6.17% attack recall), and that iterative verification recovers 158/160 episodes (59.68% Macro-F1) but leaves 59.2% (93/157) of rules matching within-seed counterexamples. We further uncover a prompt-anchoring effect across 2839 benign evaluations where zero retrieved malicious rules fire, yet unconditional rule-text injection still inflates LLM false positives from 32.19% to 39.45%.
- **Neuro-Symbolic Rule Triage Outperforming Raw Retrieval with Reduced LLM Compute:** VLEG-CR elevates verified $\mathrm{CE}_{\mathrm{seed}} = 0$ convergence from 40.0% (64/160) to 84.4% (135/160) and reduces unseen cross-seed counterexamples from 72.0% to 40.7%. Pairing CR rules with deterministic neuro-symbolic triage enables VLEG-CR-NS (64.85% Macro-F1, 95% CI $[+0.13, +1.71]$ pp) and Pilot-selected VLEG-CR-Bank (65.30% Macro-F1, 95% CI $[+0.28, +2.60]$ pp; 35.99% FPR) to significantly outperform raw retrieval (63.92% Macro-F1, 38.13% FPR), achieve the lowest FPR among all 32-flow baselines (including shallow CART at 36.53%), and cut online 70B LLM query calls by 47.1%–49.0%.

1.5 Paper Organization

Section 2 reviews episodic retrieval, local reflection, and intrusion detection baselines. Section 3 details the VLEG verification loop, the VLEG-CR counterexample compiler and inductive refinement procedure, and the VLEG-CR-NS neuro-symbolic inference architecture. Section 4 presents the symbolic sanity check, the UNSW-NB15 benchmark, factorial ablations, cost audits, and decision-tree comparisons, followed by limitations. Section 5 concludes the paper.

2 Related Work

2.1 Episodic Memory and Demonstration Retrieval

In-context learning uses demonstrations in the prompt to support task adaptation without parameter updates [1]. Retrieval-augmented generation (RAG) connects generation to external evidence retrieved for the current input [2]. LongMem extends access to past context through a memory bank and retrieval mechanism [10]. Related persistent-memory architectures including MemGPT [9], MemoryLLM [11], and HippoRAG [12] organize external experience across interaction sessions. Their memory contents and training procedures

differ, so they should not be treated as interchangeable implementations of a single memory architecture.

Retrieval studies also address demonstration selection [20, 21] and effective context length degradation [7, 8, 13], complementing broader surveys of agent memory design [14, 15]. These questions concern which evidence reaches inference, independently of whether that evidence is subsequently summarized into verbal rules.

For labeled classification tasks, raw-example retrieval offers a direct, zero-consolidation-cost control: each historical record retains its observed features and ground-truth label. It can preserve local exceptions that a compressed summary omits, although its usefulness depends on selecting relevant examples within a finite context budget. A retrieved example also does not supply an explicit verbal rule or establish that a neighboring target has the same label. Retrieval relevance, evidence coverage, and prediction correctness therefore require separate evaluation.

Our raw-memory baseline follows this evidence-preserving principle. It provides labeled historical episodes at inference without reflection-generation calls. The symbolic sanity check compares it with abstract and hybrid context, while the public-flow experiment shares label-free feature retrieval across local conditions to control example selection. This design isolates the supplied memory content and gating mechanism more closely than a comparison that changes both retrieval and context construction.

2.2 Local Reflection and Feedback

MemoryBank stores, retrieves, and updates information from past interactions [22]. Generative Agents combine an experience store with higher-level synthesized reflections [23], while Reflexion converts binary task feedback into verbal reflections retained for subsequent trials [16]. These works show several ways to use experience beyond storing raw examples. They differ in what constitutes an episode, which feedback signal is available, when memories are updated, and how stored text influences later behavior.

Related work also examines procedural skill acquisition in open-ended worlds [17], structured agentic memory networks [24], and timeline-based memory management [25]. Their different memory objectives motivate explicit controls for the few-shot classification setting studied here.

The distinction between environmental feedback and source-episode verification is especially relevant. Trial feedback identifies an unsuccessful action or prediction and motivates revision across episodes. A source-episode check instead asks whether a candidate explanation agrees with the observed historical record under a specified verification prompt. Neither mechanism, by itself, proves that the resulting explanation identifies the true causal factors across the broader data distribution. A reflection may repair its stated class label while retaining an unsupported threshold or treating an incidental attribute as a general condition.

VLEG studies a bounded per-episode admission procedure: generate a candidate from label-free attributes, check it against the labeled source, revise rejected candidates within budget K, and store an accepted reflection alongside its source evidence. Historical labels are available to the verifier, whereas target test labels are excluded from inference. By logging every intermediate revision attempt $j \in \{0, 1, 2\}$, our evaluation explicitly separates the effect of initial label disclosure in the first critique ($j = 1$) from multi-step verification enforcement ($j = 2$), as well as separating source-label agreement from cross-episode specificity.

2.3 Semantic Consolidation, Symbolic Rule Learners, and Intrusion Detection

Semantic consolidation compresses experience into global hypotheses or summaries that guide subsequent reasoning. Hypothesis-Driven Reasoning (HDR) iteratively generates and verifies hypotheses against

episodic evidence, then combines hypotheses with retrieved episodes at inference [18]. It provides a close comparison for the unit of abstraction: hypotheses induced across multiple episodes versus VLEG's local reflections attached to individual source records. Global hypotheses can express conditional relations; indeed, one background-conditioned statement represents our synthetic sanity-check rule. Heterogeneity alone therefore does not imply that global abstraction must fail, though compressing diverse network attacks into a single global summary risks severe over-generalization.

Inductive reasoning and hypothesis generation are studied in [19,26–29]; complementary work considers cognitive memory architectures [30–32], hallucination detection and verification [33,34], and iterative self-critique [35,36]. In the domain of network traffic analysis and intrusion detection, supervised deep models such as ET-BERT [3], interpretable shallow decision trees (CART), and tabular ensemble classifiers (e.g., Random Forest and GBDT/XGBoost evaluated on UNSW-NB15 [37] and CICIDS2017 [4]) provide strong baselines. More recently, Houssel et al. [5] investigated LLMs for explainable network intrusion detection, and Islam et al. [6] proposed MA-IDS, a multi-agent RAG framework with an experience library for IoT intrusion detection.

Hybrid grounding supplies both an abstraction and supporting examples at inference. This retains access to concrete evidence, but can introduce conflicting guidance when the abstraction is inaccurate or applies too broadly. Its effectiveness depends on the specificity of the reflection, not merely the presence of both memory types. Comparing raw, abstract, hybrid, and counterexample-refined neuro-symbolic execution against both interpretable decision trees and tree ensembles is therefore necessary to identify these operational trade-offs.

VLEG's contribution is a controlled diagnostic assessment of local admission checking, evidence–reflection pairing, and counterexample-guided rule refinement and neuro-symbolic triage, contextualized directly against shallow CART decision trees and Random Forest / GBDT baselines under both 32-flow few-shot and 88,699-flow full-train regimes. The public-flow evaluation also includes an admission-matched control (`Raw on admitted flows`) that retains VLEG's admission mask while omitting reflections, strictly isolating reflection content from episode selection.

3 The Proposed Method

3.1 Overview of Verified Localized Episodic Grounding (VLEG)

VLEG stores a labeled historical episode together with an accepted local reflection, and refines admitted reflections against contrary-label historical counterexamples via a deterministic predicate compiler. The implementation distinguishes generation, source verification, counterexample compilation, and inference roles. The following formulation describes their information access and stopping behavior; source acceptance alone does not guarantee correctness of the verifier or universal validity of an accepted rule across unseen episodes.

Let $\mathcal{E} = \{e_1, \ldots, e_n\}$ be historical support episodes whose labels are available during consolidation. A pure episodic memory stores those labeled episodes directly ($\mathcal{M}_{\text{raw}} = \mathcal{E}$). The symbolic sanity check uses synthetic scene tuples, while the public-flow evaluation uses deduplicated UNSW-NB15 feature records from separate historical and test partitions. Write $e_i = (x_i, y_i)$, where x_i denotes observed feature attributes and y_i the historical class label. The initial generator call receives only x_i; the verifier receives (x_i, y_i), and its critique reveals y_i when a proposal is rejected. Test labels are strictly excluded from inference in all settings.

Conversely, global semantic consolidation [18] attempts to distill $\mathcal{E}$ into a compact set of universal rules or hypotheses, $\mathcal{H}$. This can be modeled as finding a global hypothesis set that maximizes the likelihood of

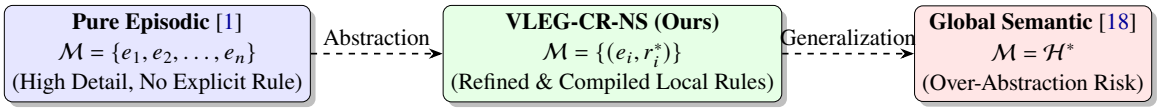


Figure 1. Theoretical positioning of VLEG and VLEG-CR-NS within the memory abstraction spectrum. VLEG bridges pure episodic retention of raw traffic and global semantic consolidation by pairing raw episodes with source-verified and counterexample-screened local abstractions.

the observed episodes:

$$\mathcal{H}^* = \arg\max_{\mathcal{H}} P(\mathcal{H} \mid e_1, e_2, \ldots, e_n). \tag{1}$$

A single conditional hypothesis can represent the contextual rule used in our symbolic sanity check. In complex network traffic datasets, however, a single global summary may omit important local exceptions or over-generalize attack conditions.

VLEG bridges this gap by shifting the abstraction target from the global distribution to the localized episode, while applying a model-based source-episode admission check (Figure 1). Rather than attempting to find a single universal $\mathcal{H}^*$, VLEG constructs a memory bank consisting of verified episodic–semantic tuples:

$$\mathcal{M}_{\text{VLEG}} = \left\{ (e_i, r_i^*) \;\middle|\; e_i \in \mathcal{E},\ \mathcal{V}(r_i^* \mid e_i) = 1 \right\}, \tag{2}$$

where r_i^* is a localized semantic reflection (a detection rule or heuristic) derived specifically from episode e_i, and $\mathcal{V}(\cdot \mid e_i)$ is a strict verification function that returns 1 when the implemented verifier accepts the candidate under its checking prompt.

The stored tuple retains the source episode alongside its accepted reflection. This permits later comparison with the concrete evidence, although the inference model may still follow an unsupported condition if the rule is overly broad.

3.2 Framework Components: Generator, Verifier, and Memory Bank

The consolidation architecture comprises a reflection generator, a strict source verifier, and a hybrid memory bank (Figure 2). Generation and checking use separate prompts and model calls to the same underlying foundation model, so shared model biases remain possible.

3.2.1 The Reflection Generator ($\mathcal{G}_\theta$)

The Reflection Generator is an LLM parameterized by θ, tasked with distilling a raw episode e_i into a localized natural-language abstraction r_i. Formally, at initial iteration $j = 0$, the generator samples a candidate reflection conditioned solely on the label-free attributes x_i:

$$r_i^{(0)} \sim \mathcal{G}_\theta(\cdot \mid x_i). \tag{3}$$

In subsequent iterations ($j > 0$), if the candidate reflection is rejected by the verifier, the generator is conditioned on the raw attributes x_i, its previous candidate reflection $r_i^{(j-1)}$, and a natural-language diagnostic critique $c_i^{(j-1)}$ provided by the verifier:

$$r_i^{(j)} \sim \mathcal{G}_\theta(\cdot \mid x_i, r_i^{(j-1)}, c_i^{(j-1)}). \tag{4}$$

The unverified single-step comparator stores its first proposal $r_i^{(0)}$ unconditionally without checking. Crucially, when $r_i^{(0)}$ is rejected due to a label mismatch, the critique $c_i^{(0)}$ explicitly informs the generator of the ground-truth label y_i; thus attempt $j = 1$ represents a label-conditioned revision, while attempt $j = 2$ represents a second verification-enforced revision if $r_i^{(1)}$ still fails verification.

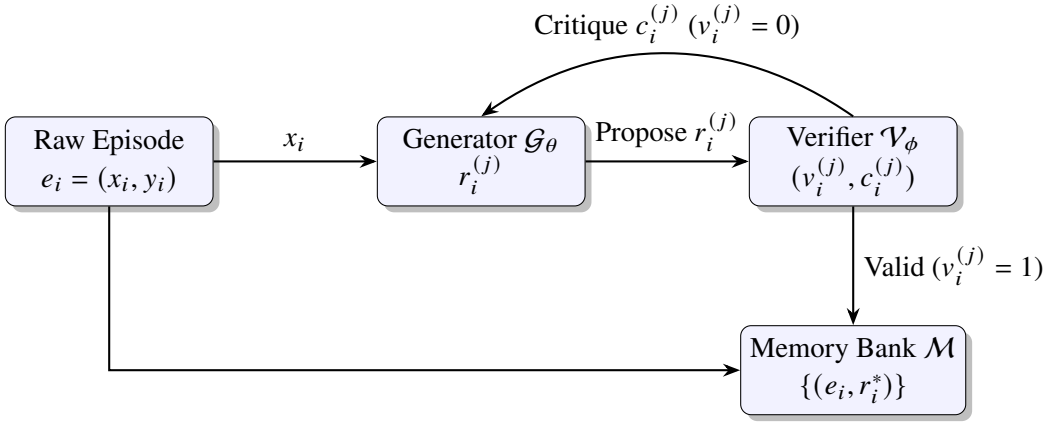


Figure 2. Architectural flow of the VLEG framework components. Separate generator and verifier calls admit only candidates accepted by the source check to the Hybrid Memory Bank alongside their raw episodic counterparts.

3.2.2 The Strict Verifier ($\mathcal{V}_\phi$)

The verifier checks the candidate reflection against its single labeled historical episode $e_i = (x_i, y_i)$. It rejects mismatched class labels, fabricated feature values, or unparseable claims. The verifier outputs a tuple consisting of a binary validity score $v_i^{(j)} \in \{0, 1\}$ and a diagnostic critique $c_i^{(j)}$:

$$(v_i^{(j)}, c_i^{(j)}) = \mathcal{V}_\phi(e_i, r_i^{(j)}). \tag{5}$$

If $v_i^{(j)} = 1$, the candidate passes the admission check. If $v_i^{(j)} = 0$, its critique $c_i^{(j)}$ is returned to the generator. Admission establishes that the verifier accepted the reflection as consistent with episode e_i; as formalized in Section 3.4, single-episode acceptance does not by itself certify cross-episode specificity.

3.2.3 The Hybrid Memory Bank ($\mathcal{M}$)

Once a reflection passes the verifier at attempt j with $v_i^{(j)} = 1$, it is denoted as the verified reflection r_i^*. The Hybrid Memory Bank $\mathcal{M}$ stores the paired tuple of the raw episode and its verified abstraction:

$$\mathcal{M} = \bigcup_{i=1}^{n} \left\{ (e_i, r_i^*) \;\middle|\; \exists\, 0 \le j < K \text{ s.t. } \mathcal{V}_\phi(e_i, r_i^{(j)})_v = 1 \right\}. \tag{6}$$

Here K is the maximum number of generation–verification attempts, indexed $j = 0, \ldots, K-1$, including the initial proposal. The first accepted candidate is retained. If none is accepted within K attempts, the episode is omitted from $\mathcal{M}$. The memory bank supports retrieval of both source evidence and a local reflection in the hybrid modality, whereas raw-only and abstract-only ablations remove one of these components for controlled comparison.

3.3 Learning Mechanism: Iterative Cyclic Refinement

The refinement loop revises a rejected candidate using the verifier critique (Figure 3). It stops at the first accepted candidate or after K attempts. Reaching the budget ceiling without acceptance omits the episode from reflection memory.

Formally, let $e_i \in \mathcal{E}$ denote a raw historical episode. At any iteration $j \in \{0, \ldots, K-1\}$, letting $(r_i^{(-1)}, c_i^{(-1)}) = (\emptyset, \emptyset)$ at $j = 0$, the generator synthesizes a candidate reflection conditioned on the label-free

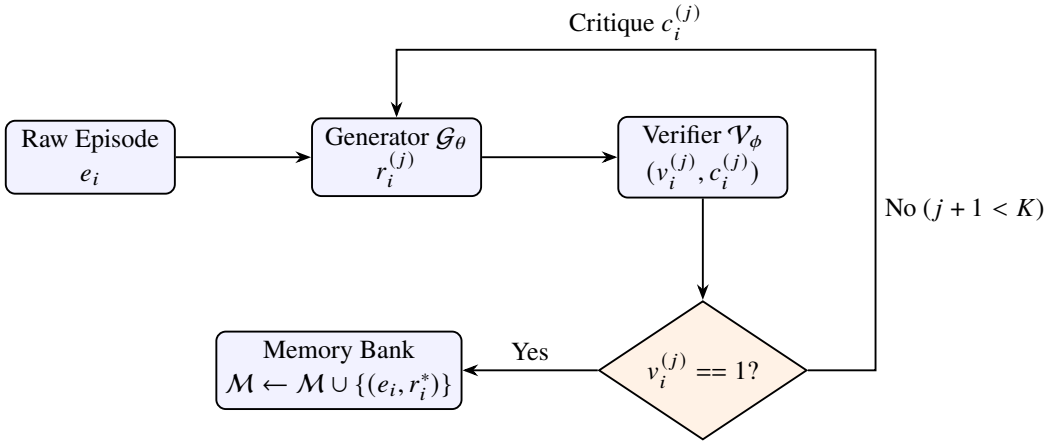


Figure 3. State transition diagram of the Iterative Cyclic Refinement mechanism. The loop stores the first candidate accepted by the verifier to the paired episodic–semantic memory bank.

attributes x_i and prior feedback:

$$r_i^{(j)} \sim \mathcal{G}_\theta(\cdot \mid x_i, r_i^{(j-1)}, c_i^{(j-1)}). \tag{7}$$

Once candidate $r_i^{(j)}$ is generated, the verifier $\mathcal{V}_\phi$ evaluates it against the full labeled source episode $e_i = (x_i, y_i)$ under a fail-closed native-Boolean JSON gate, outputting a binary validity decision $v_i^{(j)} \in \{0, 1\}$ and a diagnostic critique $c_i^{(j)}$:

$$(v_i^{(j)}, c_i^{(j)}) = \mathcal{V}_\phi\big((x_i, y_i),\ r_i^{(j)}\big). \tag{8}$$

Note that $\mathcal{V}_\phi$'s only argument besides the candidate reflection is the single episode e_i from which it was generated—by construction, the per-episode verifier has no access to any other episode in $\mathcal{E}_{-i} = \mathcal{E} \setminus \{e_i\}$. This check can expose a reflection that misstates e_i's own attributes or class label, but is not sufficient to determine whether an attribute the reflection treats as a sufficient condition is causally discriminative across episodes versus merely true of this single instance. This defines a fundamental boundary condition between single-episode source consistency and cross-episode specificity.

Define the stopping iteration j_i^* as the first attempt accepted by the verifier within the refinement budget K:

$$j_i^* = \min\left\{0 \le j < K \;\middle|\; v_i^{(j)} = 1\right\}. \tag{9}$$

If j_i^* exists, the loop terminates immediately at $j = j_i^*$ and commits $r_i^* = r_i^{(j_i^*)}$ to the memory bank; if $v_i^{(j)} = 0$ (including any malformed verifier response), the critique $c_i^{(j)}$ is fed back into the generator for iteration $j+1$.

Remark 1 (Information-Access Scope of the Per-Episode Admission Check). Let A_j denote reaching attempt j, E_j^{local} a candidate reflection that contradicts its source episode $e_i = (x_i, y_i)$ (e.g., an inverted label or fabricated feature value), and B_j its erroneous acceptance by $\mathcal{V}_\phi$. Using one-based attempt indices $j = 1, \ldots, K$, if $\Pr(E_j^{\text{local}} \mid A_j) \le p_h$ and $\Pr(B_j \mid A_j, E_j^{\text{local}}) \le p_v$, the union bound over K attempts yields:

$$\Pr(\text{local source-contradiction commitment}) \le \sum_{j=1}^{K} \Pr(A_j \cap E_j^{\text{local}} \cap B_j) \le \min\{1, K p_h p_v\}. \tag{10}$$

Crucially, Equation (10) applies only to local source contradictions E_j^{local} detectable from e_i alone. For cross-episode over-generalization errors E_j^{cross}—where r_i^* is 100% consistent with e_i but also matches contrary-label

episodes $e_m \in \mathcal{E}_{-i}$—a single-episode verifier has no discriminative signal ($\Pr(B_j \mid E_j^{\text{cross}}, \text{local agreement}) \approx 1$). Suppressing E_j^{cross} therefore requires an explicit cross-episode counterexample check across $\mathcal{E}_{-i}$.

3.4 Deterministic Counterexample Compiler and Inductive Rule Refinement (VLEG-CR)

To eliminate cross-episode over-generalization errors E_j^{cross} before rules are committed to memory, we introduce a deterministic predicate compiler Ψ and a counterexample-guided refinement stage (**VLEG-CR**).

Candidate traffic reflections follow the conjunctive form `if <cond_1> and <cond_2> ... then label = <CLASS>`. Given a candidate reflection r, compiler Ψ parses each atomic clause into an executable relational predicate over the 19 flow features:

$$\pi(r)(x) = \bigwedge_{m=1}^{d} \text{op}_m\big(x[\text{field}_m],\ \text{rhs}_m\big) \in \{0,1\}, \qquad \hat{y}(r) \in \{\texttt{BENIGN}, \texttt{MALICIOUS}\}, \tag{11}$$

where $\text{op}_m \in \{=, \neq, >, <, \geq, \leq\}$ and rhs_m is a numeric literal, categorical string literal, or feature reference (such as `sbytes > dbytes`). To prevent trivial instance memorization during refinement, Ψ rejects exact equality (=) against non-zero literals on continuous rate, duration, jitter, load, and byte-count fields, requiring inequality bounds instead.

Definition 1 (Within-Seed Counterexamples, Positive Support, and Inductive Refinement)**.** For a parseable rule r_i associated with source episode $e_i = (x_i, y_i) \in \mathcal{E}_{\text{seed}}$, let $\mathcal{E}_{-i}^{\text{seed}} = \mathcal{E}_{\text{seed}} \setminus \{e_i\}$ denote the remaining 31 historical flows in the same seed bank. Its within-seed counterexample count $\text{CE}_{\text{seed}}(r_i)$ and same-class positive support $S_{\text{pos}}(r_i)$ are defined as:

$$\text{CE}_{\text{seed}}(r_i) = \sum_{e_m \in \mathcal{E}_{-i}^{\text{seed}}} \mathbb{I}(\pi(r_i)(x_m) = 1 \wedge y_m \neq y_i)\,, \quad S_{\text{pos}}(r_i) = \sum_{e_m \in \mathcal{E}_{-i}^{\text{seed}}} \mathbb{I}(\pi(r_i)(x_m) = 1 \wedge y_m = y_i)\,. \tag{12}$$

We also define the offline cross-seed diagnostic count $\text{CE}_{\text{pool}}(r_i)$ identically over all 159 other flows across the five pooled seeds ($\mathcal{E}_{\text{all}} \setminus \{e_i\}$); CE_{pool} is never accessed during consolidation or inference.

If a Stage-A reflection r_i^* fails compilation or exhibits $\text{CE}_{\text{seed}}(r_i^*) > 0$, VLEG-CR initiates up to K_{CR} counterexample-guided refinement iterations ($t = 0, \ldots, K_{\text{CR}} - 1$). At step t, Ψ accumulates up to three falsely matched contrary-label episodes $C_i^{(t)} \subset \mathcal{E}_{-i}^{\text{seed}}$ together with univariate separating feature intervals between x_i and $C_i^{(t)}$, formatting a structured diagnostic critique $\tilde{c}_i^{(t)}$ for generator $\mathcal{G}_\theta$:

$$\tilde{r}_i^{(t)} \sim \mathcal{G}_\theta\big(\cdot \mid x_i,\ \tilde{r}_i^{(t-1)},\ \tilde{c}_i^{(t-1)}\big). \tag{13}$$

A refined candidate $\tilde{r}_i^{(t)}$ is committed as r_i^{CR} if and only if (i) Ψ verifies valid non-memorizing syntax, $\pi(\tilde{r}_i^{(t)})(x_i) = 1$, and $\text{CE}_{\text{seed}}(\tilde{r}_i^{(t)}) = 0$, and (ii) the LLM verifier confirms source fidelity $\mathcal{V}_\phi(e_i, \tilde{r}_i^{(t)})_v = 1$. If no candidate satisfies both gates within K_{CR} steps, the episode retains its raw record e_i with $r_i^{\text{CR}} = \emptyset$.

3.5 Inference Integration: Prompt Grounding vs. Neuro-Symbolic Rule Triage

Given a target flow T with features x_T, the retriever computes dense cosine similarity $S(T, \mathcal{M}_i)$ over standardized label-free feature embeddings ($z_i = x_i$ for network flows; $z_i = r_i^*$ for symbolic scenes):

$$S(T, \mathcal{M}_i) = \frac{f_{\text{emb}}(T) \cdot f_{\text{emb}}(z_i)}{\|f_{\text{emb}}(T)\|\|f_{\text{emb}}(z_i)\|}, \qquad \mathcal{M}_{\text{top}}(T) = \underset{\mathcal{S} \subseteq \mathcal{M},\, |\mathcal{S}| = \min(k, |\mathcal{M}|)}{\arg\max} \sum_{\mathcal{M}_i \in \mathcal{S}} S(T, \mathcal{M}_i). \tag{14}$$

When verified rules are injected purely as prompt text (**VLEG** and prompt-only **VLEG-CR**), even rules whose antecedent evaluates to false ($\pi(r_i)(x_T) = 0$) remain visible in the context window and can bias a quantized LLM toward predicting `MALICIOUS`. Conversely, stripping rules with $\mathrm{CE}_{\mathrm{seed}} > 0$ at the prompt level (**VLEG-CG**) avoids false alarms but forfeits explicit rule execution. Because every admitted VLEG-CR rule is already compiled into a verified predicate $\pi(r_i^{\mathrm{CR}})$ with $\mathrm{CE}_{\mathrm{seed}} = 0$, our **Neuro-Symbolic Rule Triage** executes compiled predicates deterministically on x_T before invoking LLM inference. Let $\mathcal{R}_{\mathrm{cand}}(T)$ denote a candidate rule scope and $\tau_{\mathrm{sup}} \geq 0$ a minimum positive-support threshold. Define the firing vote counts for class $y \in \{\texttt{BENIGN}, \texttt{MALICIOUS}\}$ as:

$$V_y(x_T) = \sum_{(e_i, r_i^{\mathrm{CR}}) \in \mathcal{R}_{\mathrm{cand}}(T)} \mathbb{I}\Big(r_i^{\mathrm{CR}} \neq \emptyset \ \wedge \ S_{\mathrm{pos}}(r_i^{\mathrm{CR}}) \geq \tau_{\mathrm{sup}} \ \wedge \ \pi(r_i^{\mathrm{CR}})(x_T) = 1 \ \wedge \ \hat{y}(r_i^{\mathrm{CR}}) = y\Big). \tag{15}$$

We evaluate two neuro-symbolic execution scopes that fall back to raw retrieved episodes $\{e_i\} \in \mathcal{M}_{\mathrm{top}}(T)$ via LLM_ψ only when no unambiguous rule decision is triggered:

1. **Shared-Retrieval Neuro-Symbolic Triage (VLEG-CR-NS, $\mathcal{R}_{\mathrm{cand}} = \mathcal{M}_{\mathrm{top}}(T)$, $\tau_{\mathrm{sup}} = 0$):** Evaluates compiled CR rules strictly within the same top-k retrieved episodes as raw retrieval. If $V_{\texttt{MAL}}(x_T) > 0$ and $V_{\texttt{BEN}}(x_T) = 0$ (or vice versa), it outputs the unanimous class directly with the firing human-readable rule(s) and zero LLM call; otherwise it queries LLM_ψ on raw retrieved neighbors.
2. **Support-Filtered Bank Triage (VLEG-CR-Bank, $\mathcal{R}_{\mathrm{cand}} = \mathcal{M}$, τ_{sup} selected on pilot):** Evaluates all compiled $\mathrm{CE}_{\mathrm{seed}} = 0$ rules in the 32-flow bank satisfying $S_{\mathrm{pos}}(r_i^{\mathrm{CR}}) \geq \tau_{\mathrm{sup}}$. When $V_{\texttt{MAL}}(x_T) \neq V_{\texttt{BEN}}(x_T)$, the majority class and its supporting rules are returned deterministically; ties or uncovered flows ($V_{\texttt{MAL}} = V_{\texttt{BEN}}$) fall back to top-$k$ raw-flow LLM inference.

3.6 Complete Two-Phase Algorithm

Algorithm 1 summarizes the complete two-phase VLEG-CR and VLEG-CR-NS / VLEG-CR-Bank pipeline, uniting Stage-A source verification, Stage-B counterexample-guided inductive refinement, and Phase-2 neuro-symbolic triage.

4 Experiments

4.1 Experimental Settings and Two-Stage Protocol

Our evaluation follows a two-stage protocol: (1) a controlled 16-scene symbolic sanity check (Sections 4.2–4.6) that tests per-episode verification, grounding modalities, refinement budgets, and distractor attribution against a known oracle at 7B and 70B scales, and (2) a public network-flow benchmark on deduplicated UNSW-NB15 telemetry (Sections 4.7–4.8) evaluating Stage-A source verification, Stage-B counterexample-guided refinement (VLEG-CR), prompt gating (VLEG-CG), and Phase-2 neuro-symbolic triage (VLEG-CR-NS and VLEG-CR-Bank) against shallow CART decision trees, Random Forests, and GBDT ensembles.

In the symbolic sanity check, synthetic scenes vary across shape, color, texture, background, and a deterministic size distractor. On white backgrounds, `ANOMALY` requires a red circle; black backgrounds invert this rule, while texture and size have zero causal effect. Level 1 tests eight white-background scenes; Level 2 tests all sixteen scenes across both backgrounds; Level 3 withholds four scenes from the 12-episode memory bank and tests all sixteen across multiple masking seeds. Symbolic experiments run Qwen2.5-7B-Instruct and Llama-3.3-70B-Instruct with `nomic-embed-text` embeddings at temperature 0, withholding target labels from all inference calls.

Algorithm 1 Counterexample-Refined VLEG with Neuro-Symbolic Rule Triage (VLEG-CR-NS / Bank)

Input: Support bank $\mathcal{E}_{\text{seed}}$, target $T = (x_T)$, budgets (K, K_{CR}), top-k, threshold τ_{sup}, scope $\in$ {Top-k, Bank}

Initialize Memory Bank $\mathcal{M} \leftarrow \emptyset$
for each $e_i = (x_i, y_i) \in \mathcal{E}_{\text{seed}}$ **do** ▷ Phase 1 Stage A: Source-Episode Verification
  $j \leftarrow 0$; $(r, c, v) \leftarrow (\emptyset, \emptyset, 0)$
  while $v = 0$ **and** $j < K$ **do**
    $r \leftarrow \mathcal{G}_\theta(x_i, r, c)$; $(v, c) \leftarrow \mathcal{V}_\phi(e_i, r)$; $j \leftarrow j + 1$
  end while
  $r_i^{\text{CR}} \leftarrow \emptyset$
  if $v = 1$ **and** $\Psi(r)$ valid **and** $\text{CE}_{\text{seed}}(r) = 0$ **then**
    $r_i^{\text{CR}} \leftarrow r$
  else ▷ Phase 1 Stage B: Counterexample-Guided Inductive Refinement
    for $t = 0$ **to** $K_{\text{CR}} - 1$ **do**
      Build counterexample critique $\tilde{c}$ from falsely matched contrary flows in $\mathcal{E}_{\text{seed}} \setminus \{e_i\}$
      $r \leftarrow \mathcal{G}_\theta(x_i, r, \tilde{c})$
      if $\Psi(r)$ valid **and** $\pi(r)(x_i) = 1$ **and** $\text{CE}_{\text{seed}}(r) = 0$ **and** $\mathcal{V}_\phi(e_i, r)_v = 1$ **then**
        $r_i^{\text{CR}} \leftarrow r$; **break**
      end if
    end for
  end if
  $\mathcal{M} \leftarrow \mathcal{M} \cup \{(e_i, r_i^{\text{CR}}, S_{\text{pos}}(r_i^{\text{CR}}))\}$
end for
$\mathcal{M}_{\text{top}} \leftarrow \text{Retrieve}(\mathcal{M}, x_T, k)$; $\mathcal{R}_{\text{cand}} \leftarrow \mathcal{M}_{\text{top}}$ if scope = Top-k else $\mathcal{M}$ ▷ Phase 2: Inference
Compute deterministic predicate votes $(V_{\texttt{BEN}}(x_T), V_{\texttt{MAL}}(x_T))$ over $\mathcal{R}_{\text{cand}}$ via Eq. (15)
if unambiguous rule decision ($V_{\texttt{MAL}} \neq V_{\texttt{BEN}}$, and $\min(V_{\texttt{BEN}}, V_{\texttt{MAL}}) = 0$ for Top-k) **then**
  return $\arg\max_y V_y(x_T)$ with firing verified rule rationales (0 LLM calls)
else
  return $\text{LLM}_\psi\big(\text{PromptBuilder}(\{e_i\}_{(e_i,\cdot)\in\mathcal{M}_{\text{top}}}, x_T)\big)$ ▷ Raw-retrieval LLM fallback
end if

4.2 Controlled Memory-Strategy Comparison (Symbolic Sanity Check)

On the 16-scene symbolic sanity check, verified local consolidation eliminates the label-inversion collapse of unverified reflection and reaches accuracy comparable to raw retrieval at 70B (91.67% **versus** 89.58%**,** 44/48 **versus** 43/48**), while trailing raw retrieval at 7B (**75.0% **versus** 87.5%**).** To isolate core consolidation mechanisms under identical model weights and retrieval budgets, we compare VLEG ($K = 3$) against global-hypothesis consolidation (HDR adaptation), unverified single-step reflection, and raw in-context learning (ICL) across the three symbolic complexity levels.

Within each model-scale comparison, the same underlying model and temperature (0) are used. Level 1 has eight queries and Level 2 has sixteen. Level 3 is pooled over three masking seeds, giving 48 evaluations of the sixteen recurring scene combinations. Repeat runs matched reported accuracies in the checked conditions, representing an empirical repeatability observation under greedy decoding.

As shown in Figure 4, the 7B accuracies for raw retrieval (ICL) are 100.0%, 93.8%, and 87.5% at Levels 1–3, compared with VLEG at 75.0%, 87.5%, and 75.0%. Global-hypothesis consolidation (HDR) recovers the uniform-background rule at Level 1 (100.0%), but drops to 62.5% at Level 2 and a pooled accuracy of 70.8% at Level 3 once the rule becomes background-conditional. The unverified single-step reflection comparator scores 25.0%, 50.0%, and 50.0%; inspected first proposals predominantly state `ANOMALY`, so its aggregate accuracy tracks the anomaly base rate rather than discriminating between scenes.

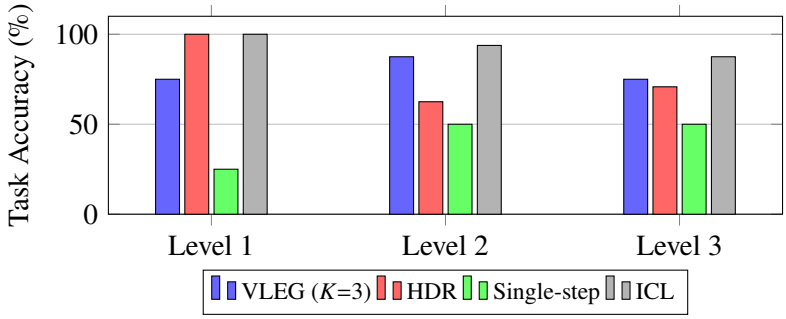


Figure 4. Measured Task Accuracy (%) across three levels of symbolic task complexity (Qwen2.5-7B-Instruct; Level 3 pooled across 3 masking seeds, $n = 48$). ICL leads at every level at 7B; HDR is bimodal (perfect at Level 1's non-conditional rule, weaker once the rule becomes background-conditional); unverified Single-step tracks each level's `ANOMALY` base rate.

At the 7B scale, VLEG outperforms unverified single-step reflection by up to +50.0 pp and exceeds global hypothesis consolidation on background-conditional rules by +25.0 pp at Level 2 and +4.2 pp at Level 3, while trailing raw retrieval.

The 70B evaluation tests Llama-3.3-70B-Instruct across the three Level-3 masking seeds ($n = 48$) and pairs VLEG against a stronger multi-attempt binary-feedback Reflexion comparator as well as ICL and HDR. At 70B, Level-3 pooled accuracies are VLEG 91.67% (44/48), raw retrieval (ICL) 89.58% (43/48), global hypothesis (HDR) 83.33% (40/48), and feedback-based Reflexion 81.25% (39/48). VLEG and raw retrieval perform comparably at 70B (differing by one query out of 48, with VLEG higher in two seeds and lower in the third). Because the model family also changes between 7B and 70B and the symbolic population is small, this shift should be interpreted as a model-dependent capability difference rather than an isolated scaling law.

4.3 Component Isolation (Grounding Modality Ablation)

Pairing verified reflections with raw episodes improves seen-rule classification at 7B (87.5%) but penalizes unseen combinations (60.0% versus 70.0%) due to distractor-attribute anchoring, a penalty that narrows at the 70B scale. To isolate inference-context composition from retrieval selection, this ablation fixes the retrieved top-k tuples using reflection-embedding similarity while varying whether the prompt displays both (e_i, r_i^*) in Hybrid Grounding, only r_i^* in Abstract-Only, or only e_i in Raw-Only [1].

At the 7B scale, Figure 5 shows that Hybrid Grounding achieves the highest accuracy on seen combinations at Level 2 (87.5% versus 81.2% for both single-modality baselines). However, on Level 3 ($n = 80$ pooled evaluations across five masks), Abstract-Only exceeds Hybrid by 5.0 pp overall (78.8% versus 73.8%, with Raw-Only at 75.0%) and by 10.0 pp on withheld masked combinations (70.0% versus 60.0%, $n = 20$). Hybrid does not outperform both alternatives in any individual 7B Level-3 seed.

At the 70B scale across the same five masks, the overall ordering reverses in favor of evidence-retaining modalities, with Hybrid and Raw-Only tying at 95.0% overall compared with 91.25% for Abstract-Only, while Abstract-Only retains a narrower 5.0 pp edge on the masked subset (85.0% versus 80.0%, with per-seed masked accuracies of 100%, 75%, 50%, 100%, and 100%).

These results separate overall performance from withheld-combination transfer. Supplying raw evidence alongside a reflection does not guarantee a benefit when the reflection embeds incidental distractor attributes (`texture` and `size`) that happen to hold in the source episode. When both e_i and r_i^* are presented for an

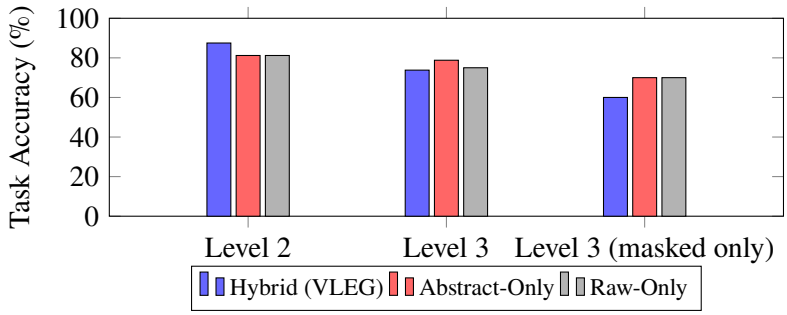


Figure 5. Measured Task Accuracy across grounding modalities (Qwen2.5-7B-Instruct). Level 3 figures are pooled across 5 masking seeds ($n = 80$ overall, $n = 20$ for the masked-only subset). Hybrid Grounding leads on Level 2 (87.5%) but trails single-modality baselines on unseen Level-3 combinations at 7B (60.0% versus 70.0%).

unseen Level-3 combination, the 7B model can over-anchor on spurious conjunctions shared between the raw example and the rule text, whereas the 70B model generates fewer distractor references during consolidation (Section 4.6).

4.4 Iterative Refinement Ablation

Bounded cyclic refinement ($K = 3$) substantially improves downstream accuracy primarily by recovering rejected historical episodes rather than by purifying causal explanations. Both $K = 1$ and $K = 3$ employ the same generator, strict verifier, and hybrid inference module, stopping at the first accepted proposal. Whereas $K = 1$ immediately discards any episode whose initial reflection fails verification, $K = 3$ allows up to two critique-guided revisions before omission.

Across five 7B masking seeds, $K = 3$ achieves 75.0% overall accuracy (60/80), compared with 63.75% (51/80) at $K = 1$, an 11.25 pp increase. Masked-only accuracy rises from 50.0% (10/20) to 60.0% (12/20). The masked-only difference is zero in three seeds and positive in two. The pooled overall gain therefore does not imply that every memory draw improves transfer to withheld combinations. These are repeated evaluations of the same sixteen-element population under different masks, not eighty independent scenes.

The principal observed change is memory admission coverage. First-attempt verification ($K = 1$) retains 42%–58% of historical episodes across seeds, whereas $K = 3$ retains 83%–100%. Rejection critique can reveal the correct historical label, allowing a revised candidate to agree with its source. Additional attempts consequently recover examples that the smaller budget would omit. The downstream improvement is consistent with this coverage recovery pathway; it is not evidence that each recovered explanation identifies the true causal attributes more reliably.

In the 70B comparison across the same five masks, increasing the budget from $K = 1$ to $K = 3$ yields a mean overall accuracy gain of 26.2 ± 8.1 pp, admitting all twelve historical episodes (100%) at $K = 3$ in every seed. Neither the pooled gain nor full admission proves universal causal generalization outside the synthetic oracle.

The comparison clarifies what refinement can and cannot change. Increasing K changes the opportunity to revise and the resulting evidence available at inference; it does not add new source observations or independently validate a generalized rule across episodes. Coverage recovery and explanation specificity must therefore be evaluated as distinct dimensions, since an accepted reflection can match its historical label while still retaining an incidental attribute. Overall, these findings validate bounded refinement as an effective mechanism for preventing rejection-induced memory starvation while motivating a joint analysis

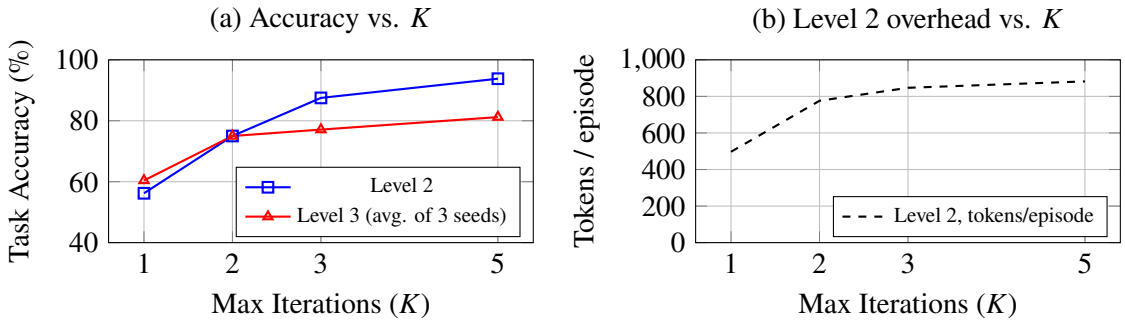


Figure 6. Measured parameter sensitivity of the maximum refinement iterations K (Qwen2.5-7B-Instruct). (a) Task Accuracy continues improving through $K = 5$ at 7B; (b) token overhead per consolidated episode is concave, with most of the cost increase occurring between $K = 1$ and $K = 2$.

of accuracy gains against token construction overhead.

4.5 Parameter Sensitivity and Overhead Analysis

Consolidation token overhead grows concavely with the refinement ceiling K because most historical episodes converge within two verification attempts. To evaluate the efficiency frontier of cyclic refinement, we sweep the maximum budget over $K \in \{1, 2, 3, 5\}$ and track both task accuracy and total prompt-plus-completion tokens per consolidated episode. At $K = 1$, verification still occurs, but a rejected proposal cannot be revised. Level 2 uses the fixed 16-scene population; Level 3 varies the withheld subset across three masking seeds.

Figure 6(a) shows no uniform 7B plateau at $K = 3$: Level-2 accuracy rises from 87.5% at $K = 3$ to 93.8% at $K = 5$, and Level-3 mean accuracy rises from 77.1% to 81.2% (with only one of the three individual masks plateauing at $K = 3$). A budget sufficient for one memory draw can therefore remain insufficient for another at 7B.

As shown in Figure 6(b), construction overhead grows concavely over the measured ceilings, rising by 279 tokens per episode from $K = 1$ to $K = 2$ (497 to 776 tokens), by 71 tokens from $K = 2$ to $K = 3$ (847 tokens), and by only 35 tokens from $K = 3$ to $K = 5$ (882 tokens). Because the vast majority of episodes pass verification on the first or second attempt, extending K invokes additional LLM calls only for the small tail of unresolved episodes.

At 70B, all three Level-3 masks yield identical admitted memories, token totals, and accuracy across $K = 2$, $K = 3$, and $K = 5$, as no historical episode requires a third attempt in those runs. The observed convergence point is therefore $K = 2$ for the 70B model on this sanity check, rather than a universal stopping rule across scales. Wall-clock timing is not used to rank these historical runs because concurrent GPU workloads affected latency, and a Level-2 repeat at $K = 3$ shifted total tokens slightly from 12,541 to 12,625 while preserving accuracy, confirming that temperature 0 does not guarantee bit-exact hardware determinism.

4.6 Diagnostic Assessment of Memory Quality

Single-episode verification enforces 100% observed source-label agreement on admitted symbolic memories across scales, whereas suppressing non-causal distractor attributes depends on foundation-model characteristics rather than local admission checking. To disentangle what per-episode verification filters from what passes through, we evaluate admitted-memory coverage, deterministic source-label consistency, and keyword references to non-causal attributes (`texture` and `size`).

Table 1. Diagnostic assessment of memory quality on the symbolic sanity check (Qwen2.5-7B-Instruct). Admission is stored episodes divided by historical episodes. Label correctness and spurious-attribution rate are computed deterministically against ground truth. "Spurious (of correct)" = % of label-correct reflections containing a texture/size keyword.

Level	Strategy	Admission	Label-Correct	Spurious (of correct)
Level 2	Unverified Single-Step	N/A (Always commit)	50.0% (8/16)	87.5% (7/8)
	VLEG (K=3)	**93.8% (15/16)**	**100.0% (15/15)**	86.7% (13/15)
Level 3	Unverified Single-Step	N/A (Always commit)	41.7% (5/12)	80.0% (4/5)
	VLEG (K=3)	**100.0% (12/12)**	**100.0% (12/12)**	83.3% (10/12)

Coverage is the proportion of historical episodes admitted within $K = 3$; it is not applicable as a filtering metric for an always-commit comparator. Source-label consistency is scored deterministically against the ground-truth scene label. The spurious-attribution flag detects references to `texture` or `size`, which have zero effect on the ground-truth label. Because a keyword flag detects any mention as well as a causal assertion, it serves as a conservative upper-bound diagnostic.

Table 1 reports 100.0% observed source-label agreement among admitted VLEG reflections at both levels (15/15 at Level 2 and 12/12 at Level 3), compared with 41.7%–50.0% for the unverified single-step comparator. Conditional on label correctness, however, unsupported-attribution rates are 86.7% versus 87.5% at Level 2 and 83.3% versus 80.0% at Level 3, showing no verification-specific reduction in distractor mentions.

In this diagnostic, admitted reflections match their source labels but frequently cite attributes that do not govern the synthetic rule. For example, asserting that a striped texture is required for an anomaly cannot be falsified by inspecting a single striped anomalous scene. Cross-episode contrast is required to separate an incidental instance property from a necessary class condition.

At 70B (Llama-3.3-70B-Instruct), aligning admitted reflections with their originating support episodes likewise yields 100.0% source-label consistency, compared with 50.0% for unverified first proposals. Unconditional texture/size keyword rates at 70B drop to 12.5% (2/16) at Level 2 and 16.7% (2/12) at Level 3 for both VLEG and unverified proposals alike (down from 83.3%–93.8% at 7B). Because this reduction occurs equally in both strategies, it reflects the underlying model's preference for parsimonious rules rather than a filtering effect of single-episode verification.

This distinction is central to interpreting any verified memory bank: admission is conditioned on a single source episode, whereas downstream classification depends on cross-episode generalization. Agreement at the first stage cannot certify specificity at the second.

4.7 Public-Flow Benchmark on UNSW-NB15: Counterexample Refinement and Neuro-Symbolic Triage

On deduplicated UNSW-NB15 telemetry, counterexample-guided refinement (VLEG-CR) raises verified zero-counterexample rule convergence from 40.0% **to** 84.4% (135/160)**, and pairing compiled CR rules with neuro-symbolic execution enables VLEG-CR-NS (**64.85% **Macro-F1) and VLEG-CR-Bank (**65.30% **Macro-F1,** 35.99% **FPR) to significantly outperform raw retrieval (**63.92% **Macro-F1,** 38.13% **FPR) while cutting online 70B LLM calls by nearly half.** Following deterministic 19-feature deduplication, conflicting-label removal, and train/test deduplication on UNSW-NB15 [37], 88,699 unique training flow fingerprints (from 175,341 raw rows) and 45,215 disjoint testing fingerprints (from 82,332 raw rows) remain. Five historical seeds (`20261020`–`20261024`) each sample 32 support flows (16 benign, 16 malicious;

Table 2. Public-flow classification performance on the 1000 shared deduplicated UNSW-NB15 test fingerprints (mean ± sample standard deviation across five 32-flow seeds). Invalid LLM outputs count as errors. All local LLM and neuro-symbolic methods use the exact same 32 support flows per seed; full-train models use all 88,699 unique training flows.

Supervision Regime	Condition / Method	Accuracy (%)	Macro-F1 (%)	Attack Recall (%)	FPR (%)
Few-Shot LLM & Neuro-Symbolic Memory (32 flows/seed, k=3, Llama-3.3-70B, 0-train)	Raw flow retrieval	65.08 ± 2.93	63.92 ± 2.54	71.39 ± 8.44	38.13 ± 6.85
	Raw on admitted flows (control)	64.86 ± 2.96	63.70 ± 2.68	71.16 ± 8.71	38.34 ± 6.61
	Unverified single-step reflection	61.38 ± 1.08	42.66 ± 1.35	6.17 ± 1.43	9.89 ± 2.03
	Global summary	50.22 ± 12.45	47.60 ± 14.13	**87.12 ± 22.27**	68.54 ± 24.20
	VLEG (Stage-A source-verified, prompt)	59.74 ± 2.03	59.68 ± 2.03	77.74 ± 5.69	48.78 ± 1.67
	VLEG-CG (within-seed CE prompt gate)	64.66 ± 2.83	63.55 ± 2.49	70.27 ± 8.14	37.86 ± 7.16
	VLEG-CR (Stage-B refined, prompt-only)	61.60 ± 2.36	61.33 ± 2.11	78.81 ± 7.62	46.82 ± 5.18
	VLEG-CR-NS (Ours, shared top-k, τ_{sup}=0)	65.78 ± 2.53	64.85 ± 2.58	74.72 ± 8.26	38.76 ± 5.17
	VLEG-CR-Bank (Ours, Pilot τ_{sup}=3)	**66.58 ± 2.79**	**65.30 ± 2.74**	71.63 ± 9.32	**35.99 ± 6.06**
Few-Shot Tabular ML (Same 32 flows/seed)	Shallow Decision Tree (CART, depth=3)	70.18 ± 1.64	69.39 ± 1.76	83.38 ± 12.86	36.53 ± 6.76
	Decision Tree (CART, unconstrained)	**70.80 ± 0.90**	**70.25 ± 0.64**	86.41 ± 8.12	37.13 ± 5.08
	Random Forest (200 trees, 32-flow)	68.86 ± 1.25	68.42 ± 1.23	**87.60 ± 10.86**	40.66 ± 5.66
	Gradient Boosting / GBDT (32-flow)	66.60 ± 4.46	65.84 ± 4.60	78.04 ± 10.27	39.22 ± 5.52
Full-Train Tabular ML (88,699 unique flows)	Shallow Decision Tree (CART, depth=5)	77.20 ± 0.00	76.81 ± 0.00	95.25 ± 0.00	31.98 ± 0.00
	Random Forest (Full-train upper bound)	**82.06 ± 0.36**	**81.31 ± 0.36**	91.99 ± 0.59	**22.99 ± 0.55**
	Gradient Boosting / GBDT (Full-train)	76.50 ± 0.00	76.25 ± 0.00	**98.22 ± 0.00**	34.54 ± 0.00

Table 3. Phase 1 consolidation and deterministic counterexample audit (Ψ) across the five formal seeds (160 historical episodes: 80 benign, 80 malicious). $CE_{seed} > 0$ is evaluated within each seed's 31 other support flows; $CE_{pool} > 0$ is evaluated offline across all 159 other flows from the five pooled seeds.

Consolidation Stage / Method	Admitted Rules	Benign Match	Malicious Match	Zero CE_{seed} Coverage	Within-Seed CE > 0	Unseen Cross-Seed CE > 0
Stage A, j=0 (Unverified single-step)	160 / 160	80 / 80 (100%)	4 / 80 (5.0%)	34 / 160 (21.3%)	126 / 160 (78.8%)	136 / 160 (85.0%)
Stage A, $j \leq 1$ (1st label-informed rev.)	95 / 160	80 / 80 (100%)	15 / 80 (18.8%)	—	—	—
Stage A, $j \leq 2$ (VLEG K=3 source-only)	158 / 160	80 / 80 (100%)	78 / 78 (100%)	64 / 160 (40.0%)	93 / 157 (59.2%)	113 / 157 (72.0%)
Stage B (VLEG-CR, K_{CR}=3 refined)	**135 / 160**	**67 / 67 (100%)**	**68 / 68 (100%)**	**135 / 160 (84.4%)**	**0 / 135 (0.0%)**	**55 / 135 (40.7%)**

160 total support flows) and evaluate the shared 1000-query test set (663 benign, 337 malicious), disjoint from the 100-query pilot set (`pilot_20261012`). All LLM experiments run on dual NVIDIA A800 GPUs using Llama-3.3-70B-Instruct (`Q4_K_M`), `nomic-embed-text` embeddings, temperature 0, $K = K_{CR} = k = 3$, and a 32,768-token window. Table 2 summarizes all few-shot LLM, neuro-symbolic, and tabular baselines.

4.7.1 Unverified Reflection Collapse and Iterative Verification Dynamics

As shown in Table 2 and Table 3, unverified single-step reflection collapses to 6.17% ± 1.43% attack recall and 42.66% Macro-F1 because the generator labels 80/80 benign flows as `BENIGN` at $j = 0$ but mislabels 76/80 (95.0%) malicious flows as `BENIGN`. Inspecting the logged verification trajectories clarifies how recovery unfolds once the verifier's critique discloses the historical label: at attempt $j = 1$, only 11/76 (14.5%) rejected malicious episodes are revised to `MALICIOUS`, whereas 65/76 (85.5%) still output `BENIGN` and are rejected a second time. At attempt $j = 2$, 63 of those 65 episodes pass verification, restoring 100% source-label agreement across 158/160 admitted flows and lifting Macro-F1 by +17.02 pp to 59.68%. Because the second critique also carries the historical label, this recovery reflects repeated label-informed critique paired with rejection gating. Yet Stage-A VLEG still trails raw retrieval (63.92%) by −4.24 pp because 59.2% (93/157) of parseable source-consistent rules match contrary-label flows within their own 32-flow seed, inflating the FPR from 38.13% to 48.78%.

Table 4. Factorial ablation separating Phase 1 counterexample refinement (CR) from Phase 2 neuro-symbolic execution (NS) across the pilot partition ($n = 100$) and the five formal seeds (5×1000 queries). Paired cluster bootstrap 95% confidence intervals compare formal Macro-F1 against zero-cost Raw flow retrieval (63.92%).

Phase 1 Rules	Phase 2 Inference Mechanism	Pilot F1	Formal Macro-F1 (%)	Formal FPR (%)	Direct Rule Used	ΔF1 vs. Raw (95% CI)
None (Raw)	LLM prompt over top-3 raw flows	54.63%	63.92 ± 2.54	38.13 ± 6.85	0.0% (0 calls saved)	Baseline (0.00 pp)
Stage-A VLEG	Unconditional hybrid LLM prompt	48.87%	59.68 ± 2.03	48.78 ± 1.67	0.0% (0 calls saved)	−4.24 [−5.91, −2.53]
Stage-A VLEG	Prompt CE gate fallback (VLEG-CG)	54.63%	63.55 ± 2.49	37.86 ± 7.16	0.0% (12.3% rules shown)	−0.37 [−1.18, +0.42]
Stage-A VLEG	Neuro-symbolic top-3 (τ_{sup}=0, no CR)	54.63%	63.99 ± 2.62	40.36 ± 6.54	20.1 ± 5.4%	+0.08 [−0.50, +0.63]
Stage-A VLEG	Neuro-symbolic bank (τ_{sup}=3, no CR)	54.63%	63.52 ± 2.92	41.30 ± 8.15	15.0 ± 8.7%	−0.40 [−0.95, +0.19]
Stage-B CR	Unconditional hybrid LLM prompt	54.78%	61.33 ± 2.11	46.82 ± 5.18	0.0% (97.1% rules shown)	−2.59 [−4.04, −1.07]
Stage-B CR	Active-firing LLM prompt gate (top-3)	55.72%	64.08 ± 2.57	40.51 ± 4.96	0.0% (47.1% rules shown)	+0.16 [−0.58, +0.93]
Stage-B CR	**VLEG-CR-NS (top-3 direct, τ_{sup}=0)**	55.72%	**64.85 ± 2.58**	38.76 ± 5.17	**47.1 ± 7.1%**	+0.93 [+0.13, +1.71]
Stage-B CR	**VLEG-CR-Bank (bank direct, τ_{sup}=0)**	58.50%	**65.11 ± 2.85**	**35.93 ± 6.27**	**68.5 ± 7.2%**	+1.19 [−0.21, +2.71]
Stage-B CR	**VLEG-CR-Bank (Pilot-selected τ_{sup}=3)**	**61.00%**	**65.30 ± 2.74**	**35.99 ± 6.06**	**49.0 ± 14.9%**	+1.38 [+0.28, +2.60]

4.7.2 Inductive Rule Refinement (VLEG-CR) and Diagnosis of Non-Firing Prompt Anchoring

Instead of discarding the 93 over-generalized rules at inference as in VLEG-CG (which leaves only 64/160 = 40.0% of episodes with usable zero-counterexample rules and retains full 3-rule prompts on just 12.3% of test queries), Stage-B counterexample-guided refinement (**VLEG-CR**) feeds matched contrary-label flows back into $\mathcal{G}_\theta$. As reported in Table 3 (Row 4), VLEG-CR raises strict $CE_{seed} = 0$ and verifier-valid rule convergence from 40.0% to 84.4% **(**135/160**:** 67/80 **benign and** 68/80 **malicious)**. Crucially, specialization within each 32-flow seed transfers to unseen support banks: offline cross-seed counterexample incidence ($CE_{pool} > 0$ across the other four unseen seeds) drops by −31.3 pp from 72.0% (113/157) to 40.7% **(**55/135**)**, while malicious rules with zero unseen cross-seed counterexamples rise from 35.1% (27/77) to 55.9% **(**38/68**)**.

When these refined rules are unconditionally concatenated into the LLM inference prompt (prompt-only **VLEG-CR**, displaying ≥ 1 verified $CE_{seed} = 0$ rule on 97.1% of queries and 2.41 rules/query on average), test Macro-F1 improves over standard VLEG from 59.68% to 61.33% (+1.65 pp), attack recall reaches 78.81%, and unparseable LLM replies drop by half (from 30 to 15 out of 5000). However, prompt-only VLEG-CR still exhibits a 46.82% FPR and trails raw retrieval (63.92%). Auditing compiler truth values on benign test queries exposes the exact mechanism: across all 2839 **benign test evaluations where zero retrieved malicious VLEG-CR rules actually fired** ($\pi(r_i^{CR})(x_T) = 0$ for every retrieved malicious tuple), prompt-only VLEG-CR still suffered a 39.45% **false positive rate**, compared with 32.19% for raw flow retrieval on the exact same 2839 evaluations (+7.26 pp). Thus, merely displaying conditional text containing `then label = MALICIOUS` in the context window anchors the 4-bit quantized 70B LLM toward false alarms even when the rule's numerical antecedent is false.

4.7.3 Factorial Ablation and Pilot Selection of Neuro-Symbolic Triage (VLEG-CR-NS and VLEG-CR-Bank)

Executing compiled $CE_{seed} = 0$ predicates deterministically via Equation (15) eliminates non-firing prompt anchoring completely. To ensure zero test-set leakage, we screened execution scopes and support thresholds $\tau_{sup} \in \{0, 1, 2, 3, 5\}$ on the disjoint 100-query pilot set (`pilot_20261012`), where raw retrieval scored 54.63% Macro-F1 (54.29% FPR), standard VLEG scored 48.87%, prompt-only VLEG-CR scored 54.78%, shared-retrieval VLEG-CR-NS ($\tau_{sup} = 0$) reached 55.72%, and bank-wide majority triage peaked at $\tau_{sup} = 3$ with 61.00% **Macro-F1** (44.29% FPR).

Table 4 and Figure 7 establish a clear synergistic interaction between Phase 1 CR consolidation and Phase 2 NS execution:

1. **Necessity of Both Stages:** Applying neuro-symbolic execution to unrefined Stage-A VLEG rules covers

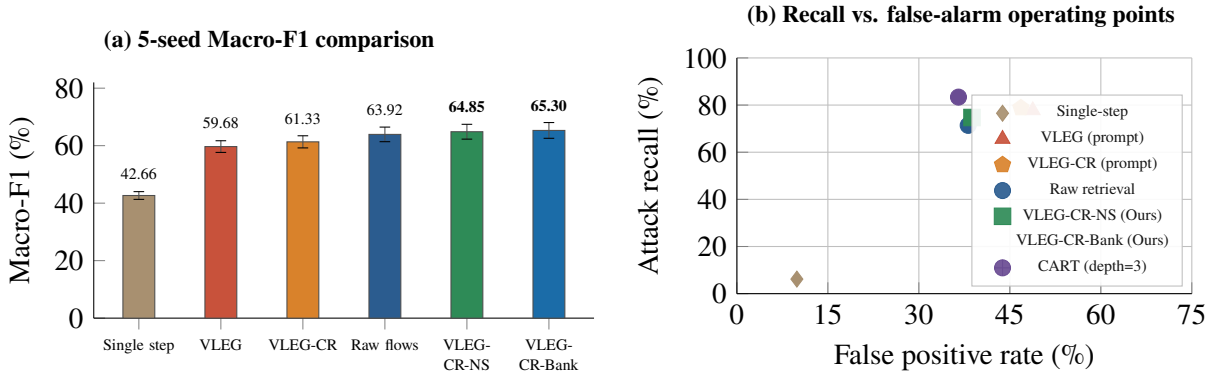


Figure 7. UNSW-NB15 five-seed performance summary. (a) Mean Macro-F1 (± SD) showing VLEG-CR-NS and VLEG-CR-Bank surpassing Raw flow retrieval; (b) attack recall vs. FPR operating points across LLM, neuro-symbolic, and shallow CART baselines.

Table 5. Memory consolidation and online per-query inference costs (5-seed means, 32 support flows and 1000 test queries per seed). VLEG-CR-NS and VLEG-CR-Bank execute compiled Python predicates in < 0.1 ms on rule-covered queries, eliminating 47.1%–49.0% of online 70B LLM calls.

Condition / Method	Consol. Calls	Consol. Tokens	Consol. (s)	LLM Calls / 1k Queries	Query Tokens / 1k	Query Mean (s)
Raw flow retrieval	0.0	0	0.0	1000.0	904,200	2.12
Unverified single-step	32.0	9405	126.5	1000.0	982,500	2.23
Standard VLEG (K=3, prompt)	120.4	40,094	419.4	1000.0	976,800	1.74
VLEG-CR (K_{CR}=3, prompt-only)	169.2	82,664	990.5	1000.0	991,400	1.81
VLEG-CR-NS (Ours, top-3 NS)	169.2	82,664	990.5	**529.0** (–47.1%)	**478,300**	**1.12**
VLEG-CR-Bank (Ours, τ_{sup}=3)	169.2	82,664	990.5	**510.0** (–49.0%)	**461,100**	**1.08**

only 15.0%–20.1% of test queries and yields no statistically significant gain over raw retrieval (+0.08 pp and −0.40 pp), because 59.2% of Stage-A rules fail the $CE_{seed} = 0$ filter. Conversely, invoking the LLM even when firing CR rules agree (`Active-firing LLM prompt gate`, 64.08%) underperforms direct symbolic execution (**VLEG-CR-NS**, 64.85%) by −0.77 pp because LLM re-interpretation reintroduces label noise.

2. **Controlled Advantage Under Identical Retrieval (VLEG-CR-NS):** Restricting rule evaluation strictly to the same top-3 retrieved episodes as raw retrieval with zero tuned hyperparameters ($\tau_{sup} = 0$), **VLEG-CR-NS** improves Macro-F1 to 64.85% ± 2.58% (+0.93 pp over raw retrieval, paired 95% CI [+0.13, +1.71] pp, outperforming raw retrieval in four of five seeds), raises attack recall by +3.33 pp to 74.72%, and classifies 47.1% of test flows directly via auditable rules at zero LLM inference cost.

3. **Lowest False-Alarm Rate via Support-Filtered Bank Triage (VLEG-CR-Bank):** Evaluating all $CE_{seed} = 0$ rules with positive support $S_{pos} \geq 3$ across the 32-flow bank (**VLEG-CR-Bank**) achieves 65.30% ± 2.74% **Macro-F1** (+1.38 pp over raw retrieval, paired 95% CI [+0.28, +2.60] pp, winning in four of five seeds), 66.58% ± 2.79% **accuracy**, and a 35.99% ± 6.06% **FPR**, while directly resolving 49.0% of test queries without calling the 70B LLM.

Table 5 reports the end-to-end consolidation and inference cost audit. Stage-B counterexample refinement adds 48.8 LLM calls and 42, 570 tokens per 32-flow seed on top of Stage-A VLEG (82, 664 total consolidation tokens). However, because VLEG-CR-NS and VLEG-CR-Bank resolve 47.1%–49.0% of incoming test queries deterministically in < 0.1 ms without invoking the 70B LLM, they save approximately 426, 000–443, 000 query tokens per 1000 test flows and cut mean per-query latency from 2.12 s (raw retrieval) to

1.08–1.12 s. Consequently, the entire one-time Phase 1 consolidation budget is amortized after fewer than 190 target flow evaluations.

4.8 Comparison with Symbolic Decision Trees (CART), Ensembles, and LLM-IDS

Benchmarking against shallow CART decision trees and tabular ensembles on the exact same splits (Table 2) clarifies the complementary strengths of classical split optimization and neuro-symbolic LLM memory:

1. **Discriminative Recall vs. False-Alarm Control in the 32-Flow Regime:** On the 32-flow splits, shallow depth-3 CART (69.39% Macro-F1, 83.38% recall, 36.53% FPR), unconstrained CART (70.25% Macro-F1, 37.13% FPR), and Random Forest (68.42% Macro-F1, 40.66% FPR) achieve higher overall Macro-F1 than VLEG-CR-Bank (65.30% Macro-F1) because global impurity minimization jointly optimizes split thresholds for recall across all 32 flows simultaneously. Remarkably, however, **VLEG-CR-Bank achieves the lowest false positive rate (**35.99%**) across all 32-flow methods**—edging out depth-3 CART (36.53%), unconstrained CART (37.13%), GBDT (39.22%), and RF (40.66%)—while matching 32-flow GBDT in overall accuracy (66.58% versus 66.60%).
2. **Relational Expressiveness and Modular Auditability:** On Seed `20261020`, the depth-3 CART tree partitions the space via three univariate axis-aligned numeric splits (`sttl <= 46.50` → `BENIGN`; `sttl > 46.50` ∧ `dinpkt <= 67.08` → `MALICIOUS`; `dinpkt > 67.08` ∧ `djit > 191.89` → `MALICIOUS`), offering no protocol semantics or cross-feature comparisons. Conversely, VLEG-CR rules express direct relational predicates across features alongside protocol rationales—for example, *"if `proto == tcp` and `state == FIN` and `dur < 1` and `sbytes > dbytes` and `sload > 50000`, then `label = MALICIOUS`, reflecting asymmetric high-rate byte transfer in a short TCP session"*—and allow SOC analysts to insert, inspect, or revoke individual episodic rules without refitting a global tree.
3. **Deduplicated Full-Train Upper Bounds:** Trained on all 88,699 unique flow fingerprints, full-train RF achieves 81.31% Macro-F1 (22.99% FPR) and GBDT achieves 76.25% Macro-F1 (34.54% FPR), confirming that removing 79,746 training duplicates and 33,161 test duplicates leaves substantial boundary overlap even for full-supervision ensembles, and situating VLEG-CR-Bank as an auditable few-shot triage layer rather than a replacement for line-rate supervised filters [5, 6].

4.9 Limitations

Several limitations bound our conclusions. First, the 16-scene symbolic sanity check tests mechanism mechanics rather than open-world network generalization, and comparing Qwen2.5-7B with Llama-3.3-70B changes model family alongside parameter count. Second, our public-flow benchmark uses 19 deduplicated tabular UNSW-NB15 features without raw packet payloads and evaluates a single quantized 70B model (`Q4_K_M`); evaluating multi-class attack families, additional traffic corpora such as CICIDS2017, and human SOC analyst triage speed remains important future work. Third, while VLEG-CR-Bank surpasses raw retrieval and achieves the lowest few-shot FPR (35.99%), a 32-flow support bank still leaves a 4.09–4.95 pp Macro-F1 gap relative to globally optimized CART trees due to lower recall on uncovered flows (51.0%) that fall back to raw LLM inference.

5 Conclusions

This paper presented Verified Localized Episodic Grounding (VLEG), Counterexample-Refined consolidation (VLEG-CR), and Neuro-Symbolic rule triage (VLEG-CR-NS and VLEG-CR-Bank) for auditable few-shot network traffic analysis. Evaluations across a 16-scene symbolic sanity check and a five-seed

benchmark on 1000 deduplicated UNSW-NB15 test fingerprints support four main conclusions:

1. **Iterative Source Verification Recovers Catastrophic Label Inversions:** Unverified single-step reflection mislabels 95.0% (76/80) of malicious support flows as benign (6.17% attack recall, 42.66% Macro-F1). Multi-step verification ($K = 3$) restores 100% source-label agreement across 158/160 admitted episodes and raises Macro-F1 to 59.68%, but leaves 59.2% (93/157) of rules matching contrary-label flows within their 32-flow seed.

2. **Counterexample-Guided Inductive Refinement Specializes Rules Across Seeds:** Feeding within-seed counterexamples back to the generator (VLEG-CR) increases strict $CE_{seed} = 0$ rule convergence from 40.0% to 84.4% (135/160) and reduces unseen cross-seed counterexamples from 72.0% to 40.7%.

3. **Neuro-Symbolic Execution Eliminates Non-Firing Prompt Anchoring and Beats Raw Retrieval:** Unconditionally injecting non-firing malicious rules into LLM prompts inflates benign false positives from 32.19% to 39.45%. Executing compiled $CE_{seed} = 0$ predicates deterministically before raw-retrieval fallback enables VLEG-CR-NS (64.85% Macro-F1, 95% CI $[+0.13, +1.71]$ pp) and Pilot-selected VLEG-CR-Bank (65.30% Macro-F1, 95% CI $[+0.28, +2.60]$ pp) to significantly outperform raw retrieval (63.92%) while cutting online 70B LLM calls and query tokens by 47.1%–49.0%.

4. **Complementary Operational Role Alongside Decision Trees:** While 32-flow CART and Random Forest models attain higher recall and Macro-F1 (68.42%–70.25%) via global split optimization, VLEG-CR-Bank achieves the lowest false positive rate (35.99%) among all 32-flow methods, matches 32-flow GBDT accuracy (66.58%), and provides cross-feature relational predicates and human-auditable protocol rationales for SOC triage.

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Availability of Data and Materials: The public network traffic benchmark evaluated in this study utilizes the publicly available UNSW-NB15 dataset [37]. All symbolic benchmark definitions, counterexample-refinement and neuro-symbolic compiler scripts, and evaluation logs supporting the reported findings are publicly available at https://github.com/creativitysurvey/VLEG.

Ethics Approval: Not applicable.

Conflicts of Interest: The authors declare no conflicts of interest to report regarding the present study.

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